



Version 1.0 — for CirclePV Limited

AI Solar Panel (PV) Inspection — India

AI Solar Inspection — Technical Proposal

AI-Powered Drone (UAV) Solar Panel Inspection

TENDER	AI Solar Panel (PV) Inspection — India
CLIENT	CirclePV Limited
PREPARED BY	RaSpect Intelligence Inspection Limited
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1. Executive Summary

CirclePV Limited operates solar PV assets that require periodic, accurate, and safe condition assessment. RaSpect proposes an **AI-powered drone (UAV) solar panel inspection** combining systematic Visual + Thermal (IR) capture with production-grade AI defect detection on the **Inspectica™** cloud platform, delivered by RaSpect's India team.

For this engagement: **2 blocks · ~11,600 m² (~1.16 MWp)**, captured by a 2-person DGCA-certified India crew over ~1–2 onsite days, producing a per-panel AI condition-assessment report (PDF) plus Inspectica platform access. Three service tiers (Standard / Advanced / Premium) are offered — see the companion Quotation.

Elevate your solar assets with insightful AI inspections. Our cutting-edge robotic (UAV) solutions revolutionise PV-asset lifecycle management, offering unparalleled safety, efficiency, and accuracy in inspection.

2. Scope of Services

1. **Onsite Drone Data Collection** — systematic Visual + Thermal (IR) capture with real-time on-site quality assurance and coverage validation.
2. **AI Data Processing** on the cloud platform **Inspectica™**, including privacy filtering and defect detection (crack, hot/cold anomaly, panel segmentation, corrosion, soiling).
3. **Annotation & Engineer Review** of each image for anomaly detection inputs and outputs (Visual and IR).
4. **Reporting** — a detailed condition-assessment report localising every defect to its panel/area.

Project parameters

Parameter	Value
Total No. of Blocks	2 Blocks
Estimated Array Area	11,600 m ² (~1.16 MWp)
Estimated Panels	~5,000–5,500
Total No. of Images	~78 Visual + ~78 Thermal (nadir grid, paired)
Onsite days	~1–2 days (team of 2: Pilot + Assistant)
Location	India

Image count is an estimate from the array area at near-nadir inspection density; confirmed post pre-site survey. Plant capacity (~1.16 MWp) is derived from the array area and to be confirmed by CirclePV.

3. Project Deliverables

5. **Per-panel condition-assessment report in PDF format** with AI defect findings for both Visual and IR imagery, localised to each panel/area on the site plan.
6. **Inspectica™ platform access** — view + query the inspection results for the term included in the chosen tier (3 / 12 / 24 months); re-processing of new data available as an extension.
7. PDF catalogue of all localised visual and thermal assets, referenced by block and area.

8. **Defect / severity register** (Advanced and Premium tiers) — every detected anomaly geo-tagged to its panel and ranked by severity and cost impact.

4. Methodology

Inspection Preparation

- 1A. Pre-Site Survey
- 1B. Specification / Job Sheet
- 1C. Drone Flight Permit (DGCA, India)

Onsite Inspection

- 2A. On-site flight prep, setup and final checks
- 2B. Drone Visual and Thermal data collection

Inspection Deliverables

- 3A. Data localisation of all captured Visual and Thermal images (by block/area)
- 3B. RaSpect **Inspectica™** platform for post-processing and visualisation (cloud)
- 3C. Defect findings summary report

Automated UAV Data Collection

Our enterprise-grade UAVs (DJI Mavic 3T — Visual + Thermal) capture the site with centimetre-level precision:

9. **Pre-Flight Planning** — automated mission planning based on panel geometry, lighting, and airspace/regulatory constraints.
10. **Systematic Image Capture** — GPS-tagged photography with overlap ensuring complete coverage.
11. **Real-Time Quality Assurance** — on-site validation of image quality, coverage gaps, and metadata before the field team departs.
12. **Secure Data Upload** — encrypted transfer to cloud processing within minutes of capture.

5. Solar Inspection AI Models

RaSpect's **Inspectica™** platform hosts 24+ AI algorithms for enterprise safety, security, and operational intelligence. Solar-relevant models include:

- **Hot Anomaly (IR)** — cell-level hotspot / over-heating detection
- **Cold Anomaly (IR)** — under-performing string / module detection
- **Panel Segmentation** — automatic panel counting and mapping
- **Crack / Corrosion / Soiling** — visual defect classification
- **Anonymity / People Masking** — privacy filtering

Production deployment accuracy: 95–98%. Edge + cloud deployment supported.

Efficiency gains vs. traditional inspection (per ~1,000-panel farm)

Stage	RaSpect	Traditional	Improvement
Autonomous Data Capture	0.5 day	2 days	+75%
AI Data Positioning	0.5 day	5 days	+90%
AI Preliminary Analysis	0.5 day	5 days	+90%

Expert Review	1 day	5 days	+80%
Reporting	0.1 day	3 days	+96%

Overall inspection efficiency improvement: ~87%.

Value proposition

- 64% more efficient workflow & teamwork via cloud platform
- 90% higher accuracy of defect prediction and localisation
- ~70% lower project cost · ~65% less time

6. Drone Third-Party Liability Insurance

India-compliant third-party UAV liability insurance (USD 5M) is included to cover third-party liability for the onsite drone operations at the India project location.

7. About RaSpect

RaSpect Intelligence Inspection Limited is a Hong Kong-headquartered AI + drone inspection company. We turn infrastructure assets into actionable data using autonomous drones, multi-sensor payloads, and production-grade AI — 1,000+ buildings across 7 countries, 3M+ images processed.

Scale	7 Countries · 1,000+ Buildings · 10,000+ Flights · 3M+ Images
Certifications	ISO 27001 · ISO 27701 · ISO 9001
Awards	KPMG PropTech 50 · Forbes Asia 100 to Watch · World Summit Awards · HK ICT Startup Gold
Platform	Inspectica 3.0 — 24+ production deep-learning defect models, digital-twin web viewer, defect register, Condition Score

8. Customer Success Stories

"Reduced inspection times by up to 50% in large industrial facilities. This led to significant cost savings and improved maintenance schedules."

"AI robotic inspections have transformed our maintenance approach — safer, faster, and more reliable."

Trusted by 100+ customers across property owners & developers, facility managers, consultants / registered engineers / building surveyors, and governments.

9. Contact

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Vision — robust technology for the safety, serviceability and sustainability (3S) of smart cities.

Mission — AI, Big Data & robotics for a cost-effective built-infrastructure safety analytics platform.

Values — Fun. Innovation. Trust. (FIT)